Varun Amitabh

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Education

- **University of Florida** — Gainesville, Florida
  Master's of Science – Electrical and Computer Engineering — *08/24 – 05/26*

  Courses: *Fundamentals of Machine Learning, Applications of Machine Learning, Pattern Recognition, Natural Language Processing, Image Processing and Computer Vision, Info Theory in ML, Deep Learning in Medical Image Analysis*

- **Symbiosis Institute of Technology** — Pune, India
  Bachelor of Technology – Electronics and Telecommunications — *09/20 – 06/24*

  Courses: *Digital Communication, Digital Image Processing, Artificial Intelligence, Machine Learning, Computer Networking, Medical Imaging, Pattern Recognition*

Skills Summary

- **Programming**: Python, C++, Go, TypeScript, Node.js, MATLAB, Linux, SQL
- **ML Frameworks & Tools**: PyTorch, TensorFlow, Transformers (HuggingFace), NumPy, Pandas, Seaborn, Keras, Scikit-learn, OpenCV, CUDA, Docker, Linux/Unix, ROS2, Gazebo, Git, Conda, TensorBoard
- **ML Skills**: Scalable ML Systems, End-to-End Model Development, Large Language Models (LLMs), Generative AI, Agentic AI Systems, Natural Language Processing (NLP), Multimodal Learning, Deep Learning, Reinforcement Learning, Representation Learning, Numerical Optimization, Statistical Modeling, Experimentation, Model Evaluation & Benchmarking
- **Other Skills**: Algorithm Optimization, Exploratory Data Analysis, Explainable AI (xAI), 3D Scene Understanding, Transfer Learning, Predictive Modeling, Memory Optimization, Data Visualisation, Multi-Sensor Fusion, A/B Testing, ML Inference, End-to-End System Building, Data & ML Pipeline Engineering, Rapid Prototyping & Debugging, Full-Stack Development, MCP Protocol & Agent Skills, API & Protocol Design, Product Ideation & Prototyping, Registry Design

Experience

- **Trustworthy Engineered Autonomy Lab** — Gainesville, Florida
  Graduate Research Assistant — *Jan 2025 – Present*
  - Developing a multimodal world model integrating LiDAR, vision and vehicle telemetry for spatial reasoning and multimodal future prediction in autonomous driving.
  - Re-engineered the world-model architecture to migrate from CARLA simulation to real-world nuScenes data, adding new spatial and KL-divergence loss terms and rebuilding data pipelines, training and evaluation end-to-end.
  - Developed modular pipelines for data ingestion, preprocessing, feature extraction and inference, enabling reproducible large-scale experimentation.
  - Implemented real-time object detection and tracking in ROS2 + Gazebo using YOLO, enabling live inference on UAV simulation feeds.
  - Debugged and stabilized system-level issues across perception, data processing, and model pipelines in a highly ambiguous research environment.
  - Optimized inference and data flow performance for real-time constraints, improving system reliability and consistency.
- **i-Heal Lab, University of Florida** — Gainesville, Florida
  Graduate Research Assistant — *Nov 2025 – Jan 2026*
  - Did the hardware/software integration for an indoor healthcare-assistant robot designed to follow clinicians and surface patient information on request.
  - Worked on the drivetrain and handled sensor mounting, electrical wiring and physical integration of onboard sensors.
  - Ingested and extracted sensor data over ROS2 to drive the robot's perception and navigation.
  - Worked toward autonomous indoor navigation by mapping floor plans and implementing path planning between locations.
- **Defense Institute of Advanced Technology (DRDO)** — Pune, India
  Research Assistant — *May 2023 – July 2023*
  - Built scalable data processing pipelines for 100GB+ spectral imaging datasets, improving downstream model performance and robustness.
  - Designed reusable preprocessing and feature engineering workflows for structured ML experimentation.
  - Developed reproducible experimentation pipelines, reducing model instability and improving evaluation consistency.
  - Applied statistical and signal processing techniques to improve data quality and downstream classification performance.
- **Electricals – Wrench Wielders Racing** — Pune, India
  Chief Technology Officer — *Nov 2022 – Dec 2023*
  - Led hands-on engineering execution at scale, managing a 25-member technical team and driving iterative debugging, prioritization, and system-level optimization.
  - Developed sensor data fusion for real-time performance monitoring.
  - Contributed to the team winning Best Overall Performer at SUPRA'23, optimizing efficiency and sensor reliability.
  - Owned end-to-end electrical system integration, from architecture decisions to on-ground debugging and deployment.

Projects

- **Agent Skills Delivery over MCP (Agentic AI)**:
  - Designed and prototyped, end-to-end, a system for delivering Agent Skills to LLM agents through MCP servers, built while skills-over-MCP was still an open, unreleased protocol question under active community discussion.
  - Served skills as MCP resources in the open SKILL.md format, letting an agent read both the skill and its attached scripts – the latter beyond the draft extension's current scope.
  - Added a digest-based manifest to support agent-side skill discovery, in line with the discovery direction the MCP working group was converging on.
  - Owned the full product arc from ideation to a shippable prototype; built in Python, Go, TypeScript and Node.js.
- **Through-Plane Multi-Image Super-Resolution on Prostate MRI**:
  - Improved diagnostic image reconstruction quality, as measured by PSNR/SSIM improvements, by implementing and benchmarking diffusion-based generative models alongside EDSR-CNN and ESRGAN architectures on the same dataset.
  - Systematically evaluated architectural trade-offs across CNN, GAN and diffusion frameworks by conducting structured ablation studies to analyze robustness and interpretability.
  - Enhanced generative model conditioning stability, as measured by reduced variance across runs, by fine-tuning diffusion models with controlled noise schedules and domain-specific feature constraints.
  - Improved model generalization by integrating large-scale augmentation, structured preprocessing and hyperparameter optimization workflows.
- **Handwritten Digit Object Detection and Classification using YOLOv5**:
  - Built a handwritten digit detection model using YOLOv5, handling 10,000+ samples of camera-captured images.
  - Preprocessed datasets and fine-tuned hyperparameters, optimizing detection speed for better results.
  - Designed deep learning-based pipelines for extracting semantic content from unstructured documents using Python and PyTorch, improving document information extraction accuracy.
  - Achieved 96% mean Average Precision (mAP), improving classification efficiency for OCR systems.

Publications

- **Deep Learning for Pulmonary Disease Identification**:
  - A. Kukker, S. Dhar, V. Singh, **V. Amitabh**, S. Likhite — 14th Intl. Conf. on Computing, Communication and Networking Technologies (ICCCNT), IEEE, 2023. DOI:10.1109/ICCCNT56998.2023.10307502
  - Built the entire CNN architecture and training/evaluation pipeline for 3-class chest X-ray classification (COVID-19 / lung opacity / normal), reaching 85% accuracy.
- **Biomedical Image Classification using Deep Reinforcement Learning**:
  - S. Parihar, A. Kukker, S. Dhar, **V. Amitabh**, V. Singh, V. Krishna — 4th Intl. Conf. on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT), IEEE, 2024. DOI:10.1109/ICAECT60202.2024.10468733
  - Built the deep-learning (CNN) and deep-RL (DQN) components for a benchmark comparing CNN, Fuzzy-CNN and Deep-RL approaches to chest X-ray classification.